COURSE

## AQA A-level Business (7138)

# Financial management: break-even

**3.1 Unit: What is business? Managing marketing and finance · 3.1.4 Financial management: break-even**

Revision for 3.1 Unit: What is business? Managing marketing and finance: each firm works it out on its own deals.

| Class | Topic |
|---|---|
| **AQA A-level Business, new specification (7138)** | **Financial management: break-even** |

THE LESSON

**Setup** — 3 min

**Round 1** — Decisions 3 min · Deals 5 min · Topic tasks 4 min

| Firm decisions | Deals | TASK 1 | TASK 2 |
|---|---|---|---|
| | | Cars to break even | Break-even limits |

**Round 2** — Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 3 min

| Firm decisions | Deals | TASK 3 | TASK 4 | B2B deal |
|---|---|---|---|---|
| | | Safety margin as % | Profit on the deal | |

**Round 3** — Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 3 min

| Firm decisions | Deals | TASK 5 | TASK 6 | B2B deal |
|---|---|---|---|---|
| | | Which price to set | Margin after rise | |

**Final questions** — 3 min

Fits a 30-minute lesson setting; longer settings add more deals.

## Each step

| | |
|---|---|
| **Firm decisions** | Each role makes its decision on its own page. |
| **Deals** | Each pupil trades with a classmate in another firm. |
| **Topic task** | The firm answers together. Everyone presses Agree. |
| **B2B deal** | One firm sells a lot of cars to another firm. |
| **Final questions** | Each pupil answers six questions. |

## Firm decisions

| | | |
|---|---|---|
| **CEO** | **Staff** | Hire staff. Each costs a salary and gives one benefit. |
| | **Sabotage** FROM ROUND 2 | Hit a rival firm's cash, at a cost. A Legal Counsel blocks it. |
| **Sales & Marketing** | **Marketing** | Book one advert campaign to lift the next sales. |
| **Finance** | **Capital & Loans** | Borrow cash. The firm pays it back with interest. |
| | **Investments** FROM ROUND 2 | Buy property or funds. The value can rise or fall. |
| **Operations** | **Seasonal stock** | Order the season's cars in advance. |

**B2B deal · Rounds 2 and 3**

One firm sells a lot of 3 cars to another firm (4 in Round 3). With an odd number of firms, two firms offer to sell and the lower price wins.

# Topic tasks

Each firm gets its own figures. The method is the same for every firm.

## Task 1

ROUND 1

They take up to 5 cars.

**Cars to break even? Round up.**

WORK OUT THE NUMBER · EXAMPLE FIGURES

| | |
|---|---|
| Livery, one-off | **£2,400** |
| Staff training, one-off | **£2,150** |
| Customer pays per car | **£18,760** |
| You pay per car | **£16,800** |

METHOD

**Break-even = (both one-off costs) ÷ (customer pays – you pay), rounded up**

(£2,400 + £2,150) ÷ £1,960 = 2.32 → 3 cars

**Why:** Part of a car does not sell: 3 whole cars cover the costs.

## Task 2

ROUND 1

You plan a fleet deal with a break-even chart.

**Which two are limits of break-even analysis?**

PICK TWO

- ✔ **It assumes every car is sold**
- ✔ **It assumes one fixed price**
- ○ It cannot show the profit
- ○ It leaves out the fixed costs
- ○ It needs the firm's cash balance

**Why:** It assumes all output sells at one price; it does show fixed costs and profit.

## Task 3

ROUND 2

Three orders, each with its break-even point.

**Order by margin of safety as a %, lowest first?**

PUT THESE IN ORDER, FIRST TO LAST · RIGHT ORDER

1. **Order 20 cars, break-even 16**
2. **Order 8 cars, break-even 6**
3. **Order 10 cars, break-even 7**

**Why:** (20 − 16) ÷ 20 = 20%, (8 − 6) ÷ 8 = 25%, (10 − 7) ÷ 10 = 30%.

## Task 4

ROUND 2

A deal with a set-up cost.

**Profit on the deal?**

WORK OUT THE NUMBER, THEN PICK IT · EXAMPLE FIGURES

| | |
|---|---|
| Livery and set-up, one-off | **£6,550** |
| Customer pays per car | **£12,475** |
| You pay per car | **£10,900** |
| They order | **9 cars** |

METHOD

**Profit = cars ordered × (customer pays − you pay) − set-up**

9 × (£12,475 − £10,900) − £6,550 = £7,625

**Why:** The first cars pay the set-up; the rest is profit: £7,625.

## Task 5

ROUND 3

You expect 7 sales at either price. At £9,500 a car you break even at 6; at £9,000, at 8.

**Which price should you set, and why?**

PICK A OR B AND GIVE A REASON

- ○ £9,000 a car
- ✔ **£9,500 a car**

**Why:** At 7 sales £9,500 clears its break-even of 6; £9,000 falls short of 8.

## Task 6

ROUND 3

They order 8 cars.

**Margin of safety now, in cars? Round break-even up.**

WORK OUT THE NUMBER · EXAMPLE FIGURES

| | |
|---|---|
| Livery and set-up, one-off | **£7,900** |
| Customer pays per car | **£16,665** |
| Cost per car before the rise | **£13,150** |
| Cost rise | **10%** |

METHOD

**Margin of safety = cars ordered − new break-even (cost after the rise)**

£13,150 + 10% = £14,465; £7,900 ÷ £2,200 → 4; 8 − 4 = 4

**Why:** The cost rise ate into the margin of safety: 4 cars left.

# Final questions

**6**
questions for each pupil

**109**
Level 3 Business questions

**8**
of them on financial management: break-even

**The game draws 6 of the 109 Level 3 questions; these 8 are about financial management: break-even.**

1 A component sells for £48. Variable cost is £30 a unit and fixed costs are £54,000 a month. What is monthly break-even output?
**3,000 units.**

2 A repair plan costs £36 a month, variable cost £12, fixed costs £72,000. Expected sales are 4,000 plans. What is the margin of safety?
**1,000 plans.**

3 A vehicle sells for £22,500 and its variable acquisition and preparation cost is £17,100. What is unit contribution before fixed costs?
**£5,400.**

4 Monthly fixed costs rise from £90,000 to £108,000 while contribution remains £300 per vehicle. By how many vehicles does break-even output increase?
**60 vehicles.**

5 Contribution is £2,400 a vehicle at today's price; matching a rival cuts it to £1,500. Fixed costs are unchanged. Which effect follows?
**Break-even rises: each sale covers less fixed cost.**

6 A tyre service charges £120 per job, variable cost is £72 and monthly fixed costs are £38,400. What is break-even output?
**800 jobs.**

7 A service plan sells for £50 a month, has variable fulfilment cost of £18 and fixed monthly support cost of £96,000. What is break-even membership?
**3,000 members.**

# Final questions

8 Fixed costs are £84,000. Each vehicle sells for £16,000 and has a variable cost of £12,000. What is the break-even output?

**21 vehicles.**